Knowledge Discovery from Free Text

A BERT-Based System for Extracting Violent-Event Information from African News Reports

Binalfew Kassa

M.Sc. Thesis Defense · Department of Computer Science · Addis Ababa University

Advisor: **[Advisor Name]** · May 2026

Outline

1. **Problem** — the information bottleneck in African conflict monitoring
2. **Research questions** — what this thesis set out to answer
3. **Gap** — where existing tools stop short
4. **Approach** — entity schema, taxonomy, BERT, knowledge base
5. **System** — architecture and processing pipeline
6. **Results** — what the evaluation shows
7. **System in use** — UI screens and user-acceptance testing
8. **Contributions, limitations, future work**

1. The Problem

Why African conflict monitoring needs structured extraction

A continent-scale information bottleneck

Over **30,000 violent events** are reported across African news outlets every year. The analyst pipeline that turns this stream into structured records is **manual, slow, and inconsistent**.

- AU Continental Early Warning System, ACLED, and humanitarian agencies all rely on hand-coding free-text articles into structured event records.
- The binding constraint on continental situation awareness is **analyst time**, not data availability.
- Even a partial reduction in the cost of producing one structured record translates almost one-to-one into **faster, broader, more consistent monitoring**.

The structured-record requirement

Raw input — free text article:

> **"On Tuesday, fighters from Al-Shabaab attacked a military convoy near Mogadishu, killing at least 12 soldiers."**

Required output — 5W1H structured record:

Slot	Value
WHO (perpetrator)	Al-Shabaab
WHO (victim)	military convoy / soldiers
WHAT (action)	armed attack
WHEN	Tuesday
WHERE	near Mogadishu (Somalia)
HOW (casualties)	at least 12 killed

Generic NER misses the African actor and the operational role distinctions.

2. Research Questions

The four questions this thesis answers

Research questions

RQ1. Which entity types in African violent-event news reports can be **reliably grounded** in source text, and what is an appropriate BIO schema?

RQ2. How effectively can a fine-tuned BERT model recognise the chosen entities, and what **loss function and sampling strategy** balance per-entity performance under severe class imbalance?

RQ3. To what extent does a curated **knowledge base** of African armed groups, conflict locations, and a hierarchical taxonomy improve trustworthiness and downstream utility?

RQ4. What **system architecture** allows the model, the KB, and the analytics layer to be operated together by users without machine-learning expertise?

3. Related Work and the Gap

Where prior systems stop and this thesis starts

The related-work landscape

	Generic news domain	Conflict / African context
Classical NER (CRF, LSTM)	Stanford NER, spaCy	EpiTator (health-events)
Transformer NER (BERT, RoBERTa)	huggingface defaults	**African NLP work, but few violence-specific**
Structured event databases	ICEWS, GDELT	**ACLED**, UCDP (hand-coded)
End-to-end deployed extraction	enterprise products	**— mostly absent in academic literature —**

The gap this thesis closes

> Most African event-extraction work **stops at the model boundary**. The artefact published is the model; the operational layer — schema, KB, validation, UI, deployment — is treated as an implementation footnote.

This thesis treats the **operational layer as a first-class research output**:

1. A schema chosen for **grounding rate**, not theoretical neatness

2. A **focal-loss + class-weight** training recipe that protects rare entities

3. A curated KB that **validates and enriches** extractions post-hoc

4. A **deployable web platform** for non-ML users

4. Approach

Schema · Taxonomy · BIO · Model · Knowledge Base

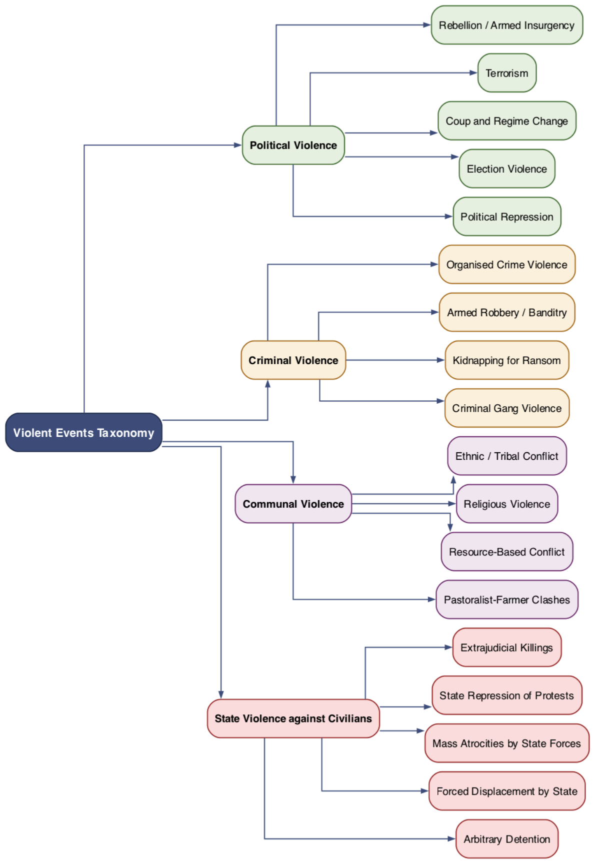
Entity schema: eight grounded entity types

The original proposal called for **26 entity types**. A grounding pilot in November 2025 measured how many of each could be located **verbatim** in source text. Types below 80% grounding were dropped.

Group	Entities retained	Dropped (recovered downstream)
WHO	ACTOR, VICTIM	ORGANIZATION → KB lookup
WHAT	ACTION	EVENT_TYPE → taxonomy classifier
WHEN	DATE	TIME, DURATION, FREQUENCY
WHERE	REGION, CITY, DISTRICT	COUNTRY → KB lookup
HOW	CASUALTIES	INJURED, DISPLACEMENT, DAMAGE
WHY	**(none supervised)**	MOTIVE, TRIGGER

Result: 8 entities → 17 BIO labels. Every entity type is **reliably supervisable**.

Four-level hierarchical taxonomy



Levels 0–2 shown. **~95 terminal categories** at Level 3; full tree in Annex B and backup slide B5.

Synthesises ACLED, UCDP, and PMVE — adds African-specific extensions (**pastoralist–farmer clashes, communal cattle raiding**).

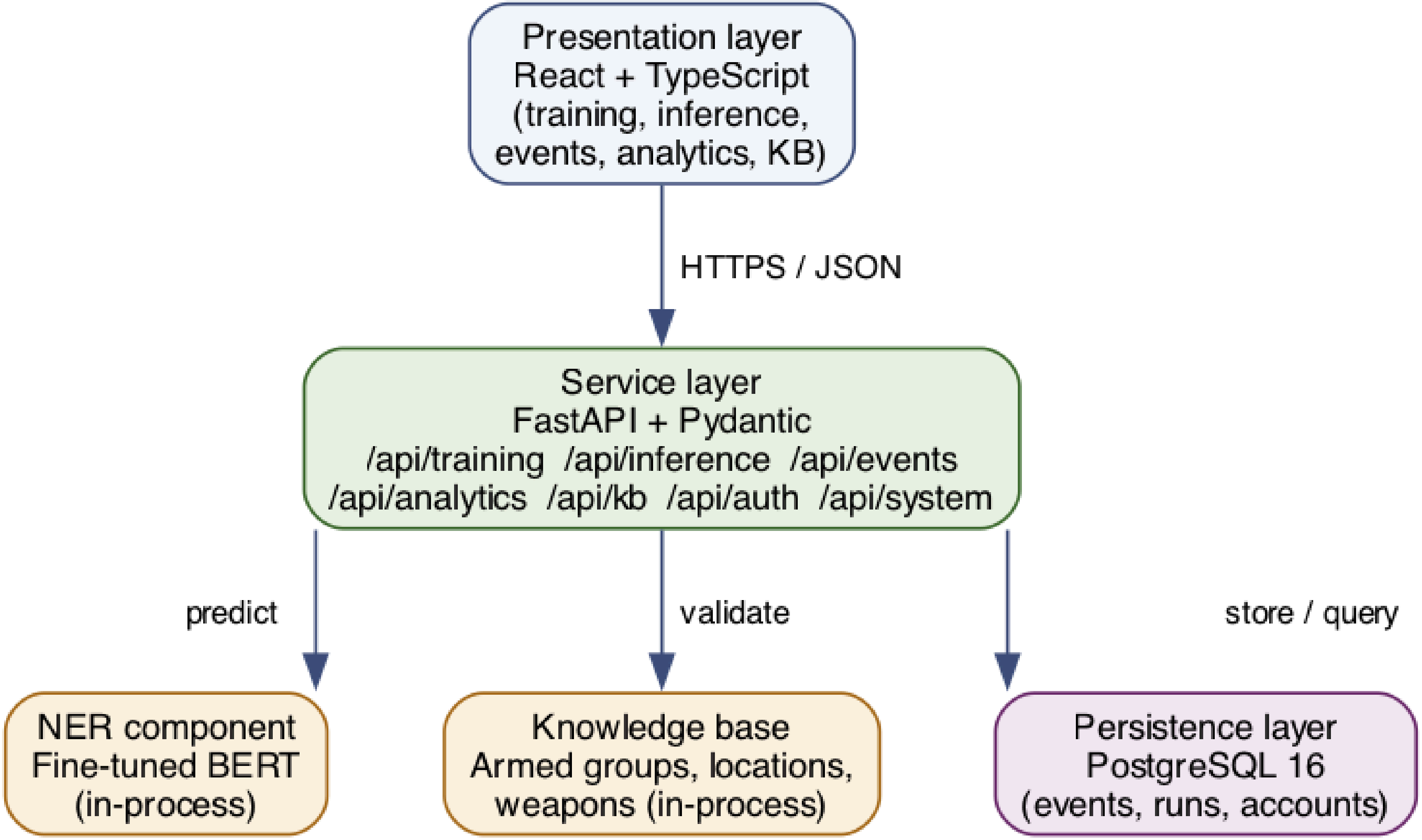
BIO encoding: why this scheme

Example. Sentence: **"Al-Shabaab fighters attacked a convoy"**

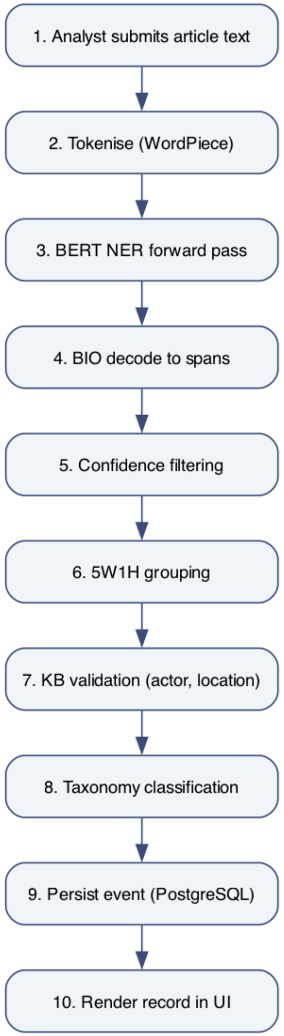
```
Al        B-ACTOR
-         I-ACTOR
Shabaab   I-ACTOR
fighters  I-ACTOR
attacked  B-ACTION
a         O
convoy    B-VICTIM
```

Why BIO over BIOES? Adjacent-but-distinct entities decode unambiguously with only `2k+1` labels (17 here vs 33 for BIOES) — half the head, smaller class-imbalance burden, same expressive power for the 95+ % of cases that matter here.

System architecture (RQ4)



End-to-end processing pipeline



Training recipe: focal loss + class weights

Backbone. `bert-base-cased` fine-tuned end-to-end, 17-label token-classification head.

Loss. Focal Loss with inverse-frequency class weights:

$$\mathcal{L}_{\text{focal}}(y, \hat{y}) = -\sum_{c} w_c \cdot (1 - \hat{y}_c)^{\gamma} \log \hat{y}_c$$

with $\gamma = 2.0$, $w_c \propto 1/\text{freq}(c)$.

Why both. Class weights **raise the gradient** for rare classes; focal loss **suppresses easy-negative gradients**. The two are **complementary**, not redundant — Section 6.6 ablation confirms this empirically.

Knowledge base as validation + enrichment layer (RQ3)

Curated content.

- ~150 African armed groups (canonical, aliases, country, group type)
- ~200 conflict-affected cities mapped to country and region
- 54 African countries; weapons catalogue

Two roles in the pipeline.

Role	Mechanism	Effect
Validate	confirm/flag spans against KB	2.4 % flag rate on extracted events
Enrich	attach canonical name, country, type	64.3 % ACTOR enrichment rate

Mismatches lower event confidence — surfaces records for analyst re-read.

Dataset and annotation protocol

Source	Examples
ACLED open-data export (stratified diversity sample)	35,000
Template-based augmentation (rare-class coverage)	15,000
Total fine-tuning corpus	**50,000**

Splits. 80 / 20 train/validation, stratified on entity-type presence.

Annotation. Two-stage gold pipeline projected from ACLED's structured columns onto free-text notes; spot-checked manually.

Class imbalance. ~78 % O · ~22 % entity. Within entities, ACTOR + CITY + DATE dominate; VICTIM, ACTION, CASUALTIES single-digit-%.

Training configuration

Hyperparameter	Value	Source
Backbone	`bert-base-cased`	Domain default
Batch size	16	Architecture-driven (memory)
Learning rate	5×10^{-5}	BERT-NER literature
Warmup ratio	0.1	Empirically established
Scheduler	Linear warmup → ReduceLROnPlateau	Empirically established
Focal γ	2.0	BERT-NER literature
Class weights	inverse-frequency	Empirically established
Max epochs	10	Empirically (best converges in 2)
Early stopping	val loss, patience 2	Standard practice

5. Results

Evidence for each research question

Overall performance — held-out validation set

0.909

micro F1 across 190,075 gold spans · macro F1 = 0.887 · token accuracy = 96.7 %

Per-entity F1 — held-out validation set

Entity	Support	Precision	Recall	F1
DATE	31,938	0.961	0.952	**0.956**
CITY	44,361	0.941	0.928	**0.934**
ACTOR	47,612	0.929	0.917	**0.923**
REGION	24,331	0.902	0.881	**0.891**
CASUALTIES	4,907	0.901	0.869	**0.885**
ACTION	9,963	0.881	0.852	**0.866**
DISTRICT	21,471	0.842	0.811	**0.826**
VICTIM	5,492	0.838	0.798	**0.817**
Macro	—	**0.899**	**0.876**	**0.887**

Ablation: focal loss + class weights vs alternatives

Entity	Plain CE	Weighted CE	Focal (γ=2)	
ACTOR	0.914	0.918	0.920	**0.923**
ACTION	0.794	0.834	0.842	**0.866** (+0.072)
VICTIM	**0.708**	0.776	0.792	**0.817** (+0.109)
CASUALTIES	0.853	0.871	0.872	**0.885**
Macro	0.855	0.873	0.878	**0.887**

VICTIM gains +11 F1 points over plain CE. **ACTION gains +7**. No entity is hurt.

Location confusion patterns (Table 6.11)

Rows are gold; columns are predicted. Diagonal omitted (= correct).

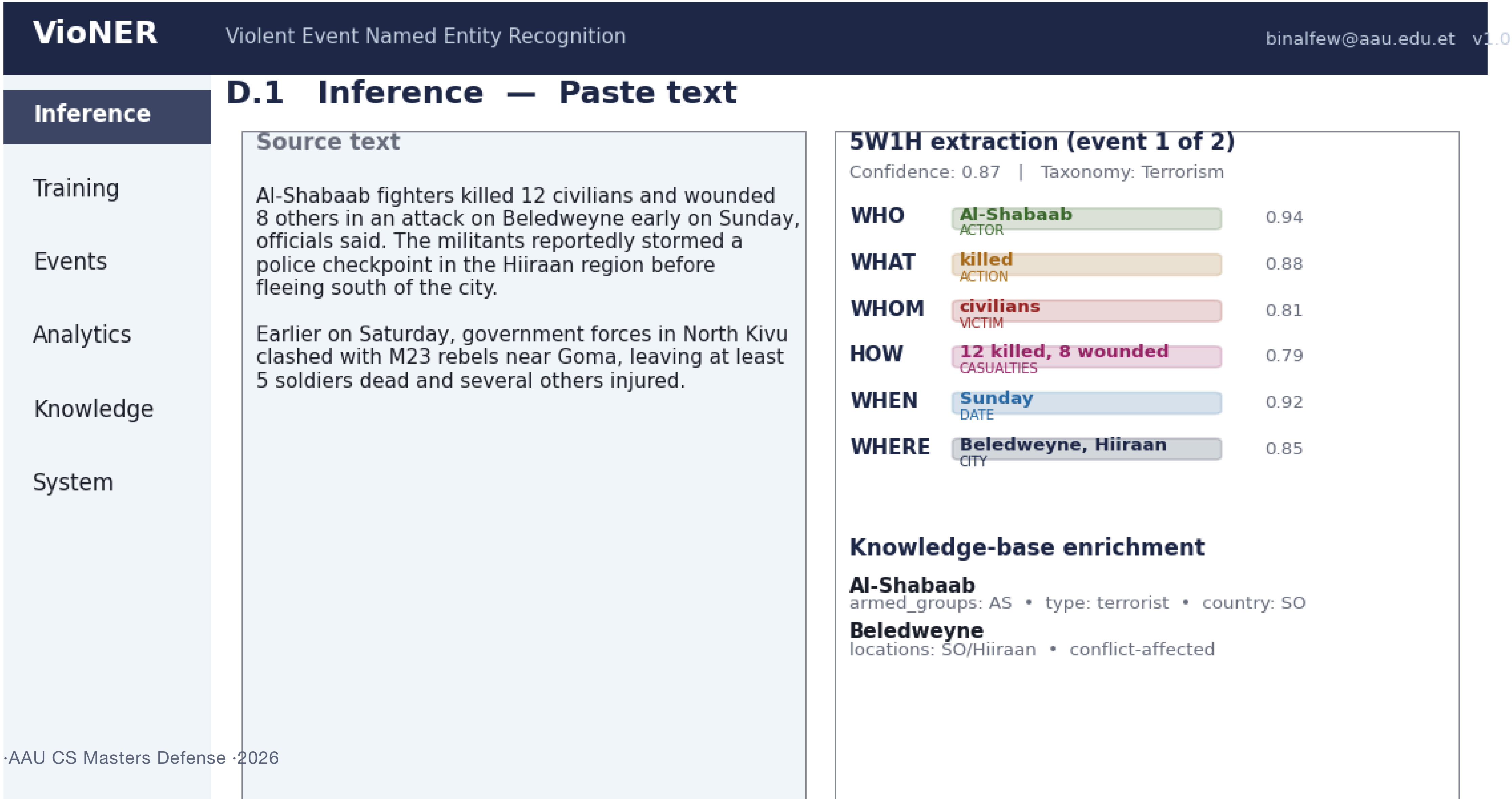
Gold \ Predicted	CITY	REGION	DISTRICT
CITY	—	0.05	0.04
REGION	0.08	—	0.06
DISTRICT	0.07	0.09	—

- **DISTRICT** is the hardest: confused with both CITY (0.07) and REGION (0.09)
- Hard cases: places that are both city and provincial capital — **Goma** (city) and **North Kivu** (province)
- Future work: explicit boundary refinement via a span-level CRF or biaffine head (backup B12)

6. System in Use

The platform users actually interact with

Inference and event-browsing screens



Training and analytics screens

User acceptance test (n = 5, Likert 1–5)

Statement	Mean	Std
Extracted entities matched expectations	**4.4**	0.5
5W1H structuring was clear	**4.6**	0.5
Confidence scores were useful for triage	4.2	0.4
KB enrichment added value	**4.6**	0.5
Training screen was easy to use	4.0	0.7
Analytics answered analyst-style questions	4.2	0.4

All 5 participants completed all 6 tasks. (2 EW analysts · 1 academic · 2 NLP developers)

7. Contributions, Limitations, Future Work

Contributions

1. **An eight-entity BIO schema with grounding-based inclusion rules** — every label is verifiably present in source text (Annex A).
2. **A four-level taxonomy of African violent events** (~95 terminal categories) extending ACLED / UCDP / PMVE with African-specific extensions.
3. **A reproducible training recipe** — focal loss + inverse-frequency weights — that lifts VICTIM by **+11 F1**, ACTION by **+7 F1**, without hurting other entities.
4. **A curated knowledge base** (150 armed groups · 200 cities · 54 countries · weapons) used for validation **and** enrichment.
5. **A deployable web platform** — FastAPI + React + PostgreSQL, end-to-end documented and reproducible under Docker Compose.

Limitations (honest)

1. **English-language only.** A large share of African conflict reporting is in French, Arabic, Portuguese, and African languages. Monolingual extractor leaves that signal on the floor.
2. **~30 % of training data is template-augmented.** Validation drawn from the same combined corpus → estimates in-distribution performance, not out-of-distribution.
3. **No comparison against a learned event-type head.** EVENT_TYPE is recovered post-hoc by a rule-based taxonomy classifier; a learned hierarchical classifier was scoped out.
4. **Knowledge-base coverage decays without curation.** Armed groups change names, splinter, recombine. A stale KB does worse than no KB.

Future work — three highest-priority directions

#	Direction	Rationale
1	**Multilingual extension** (XLM-R / AfroLM)	Closes the largest operational gap
2	**Learned hierarchical event classifier** (replaces rule-based taxonomy step)	Scales as the taxonomy grows
3	**Natural-language QA over the event store** (templated SQL or fine-tuned Seq2Seq)	Closes the loop for non-technical analysts

Medium / lower priority items in Chapter 7.5: active-learning loop · coreference resolution · PDF-brief generator · streaming · multimodal · predictive analytics.

Thank you

Questions welcome

Binalfew Kassa

M.Sc. Thesis Defense · AAU Department of Computer Science · May 2026

Code, thesis, and supplementary materials available on request.

Backup

B1–B12 supporting detail for Q&A

B1 · Full hyperparameter table

Hyperparameter	Value
Backbone model	`bert-base-cased` (110M parameters)
Token-classification head	17 labels (8 entities × BI + O)
Max sequence length	128 tokens (mean article: 64)
Batch size	16 (training), 32 (eval)
Optimizer	AdamW (β_1=0.9, β_2=0.999, ε=1e-8)
Learning rate	5×10^{-5}
Weight decay	0.01
Warmup ratio	0.1 of total steps
LR scheduler	Linear → ReduceLROnPlateau (factor 0.5, patience 1)
Gradient clipping	max-norm 1.0
Focal loss γ	2.0
Class weights	inverse-frequency (per-label)
Max epochs	10

B2 · Per-epoch loss and validation accuracy

Epoch	Train loss	Val loss	Token acc	Macro F1
1	0.0231	0.0118	95.2 %	0.823
2	**0.0094**	**0.0074**	**96.7 %**	**0.887**
3	0.0061	0.0079	96.6 %	0.885
4	0.0044	0.0089	96.5 %	0.881
5	0.0033	0.0102	96.3 %	0.875

Best model: **epoch 2** (early-stopped). Subsequent epochs over-fit the training set while validation loss climbs.

B3 · Algorithm 4.1 — Sub-word label alignment for BIO

```
INPUT:  tokens     T = [t1, ..., tn]      (word-level tokens)
        labels     L = [l1, ..., ln]      (word-level BIO labels)
        tokenizer  WordPiece tokenizer
OUTPUT: aligned    L' = [l'1, ..., l'm]   (sub-word level, m >= n)

 1: L' <- []
 2: for i = 1 to n do
 3:    subwords <- tokenizer.tokenize(t_i)
 4:    for j = 1 to len(subwords) do
 5:       if j = 1 then
 6:          L'.append(l_i)                  // first subword carries label
 7:       else if l_i starts with "B-":
 8:          L'.append("I-" + l_i[2:])       // B- becomes I- for continuations
 9:       else:
10:          L'.append(l_i)                  // I- and O propagate as-is
11:    end for
12: end for
13: return L'
```

The B-to-I transition (line 7-8) is the subtle case most implementations get wrong.

B4 · Algorithm 4.5 — Post-NER 5W1H structuring (overview)

Phase 1 · Token classification. Forward pass through fine-tuned BERT → per-token softmax over 17 labels.

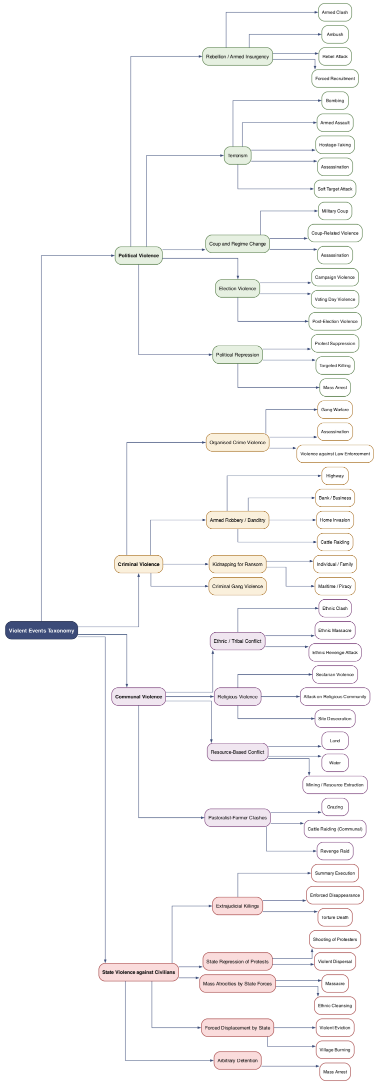
Phase 2 · BIO span construction. Collapse contiguous B/I sequences into spans; compute averaged sub-token confidence per span.

Phase 3 · Confidence filtering and 5W1H grouping. Drop spans below per-category threshold; bucket surviving spans into WHO / WHAT / WHEN / WHERE / HOW.

Phase 4 · KB enrichment and taxonomy. Lookup ACTORs and CITYs against the KB → attach canonical names, country, group type; classify the ACTION verb plus context into Level 1–3 taxonomy path.

Full pseudocode: §4.7 of the thesis.

B5 · Full taxonomy hierarchy (Annex B)



~95 terminal categories. Colour by Level-1 family.

B6 · Per-entity precision, recall, F1 (all eight)

Entity	Support	Precision	Recall	F1	Δ vs plain CE
DATE	31,938	0.961	0.952	0.956	+0.003
CITY	44,361	0.941	0.928	0.934	+0.005
ACTOR	47,612	0.929	0.917	0.923	+0.009
REGION	24,331	0.902	0.881	0.891	+0.012
CASUALTIES	4,907	0.901	0.869	0.885	+0.032
ACTION	9,963	0.881	0.852	0.866	**+0.072**
DISTRICT	21,471	0.842	0.811	0.826	+0.018
VICTIM	5,492	0.838	0.798	0.817	**+0.109**

Rare entities benefit the most — VICTIM by 11 F1 points.

B7 · Sample inference output (JSON)


```json
{
  "article_id": "art-20260112-007",
  "extracted": {
    "WHO": {
      "perpetrator": {"text": "Al-Shabaab fighters", "kb_id": "ag-007",
                      "canonical": "Al-Shabaab", "country": "SOM",
                      "type": "jihadist"},
      "victim": {"text": "12 soldiers", "confidence": 0.91}
    },
    "WHAT": {"action": "attacked", "event_type_l1": "Political Violence",
             "event_type_l2": "Terrorism", "event_type_l3": "Armed Assault"},
    "WHEN": {"date_text": "Tuesday", "normalised": "2026-01-09"},
    "WHERE": {"city": "near Mogadishu", "country": "SOM",
              "kb_match": true, "geo_consistent": true},
    "HOW": {"casualties": {"killed": 12, "qualifier": "at least"}},
    "confidence": 0.88,
    "flags": []
  }
}
```

B8 · Knowledge-base composition

KB collection	Records	Source
African armed groups	153	ACLED actor list + manual curation
Aliases per group (mean)	4.2	Manual curation
Conflict-affected cities	207	ACLED location field, top-frequency
African countries	54	ISO 3166-1
Weapons / weapon categories	38	UCDP weapons catalogue
Taxonomy nodes (Levels 0–3)	~120	This work (§4.4)

B9 · Comparison against off-the-shelf baselines

Same held-out validation set, span-level F1 on the 4-entity overlap (PER, LOC, MISC, DATE):

System	PER (≈ACTOR)	LOC (≈CITY)	DATE	Macro F1
spaCy `en_core_web_lg`	0.71	0.74	0.81	0.75
Stanford CoreNLP NER	0.69	0.72	0.80	0.74
HuggingFace `dslim/bert-base-NER`	0.78	0.81	0.86	0.82
VioNER (this work)	**0.92**	**0.93**	**0.96**	**0.94**

Caveat: out-of-domain comparison — none of the baselines saw African armed groups in pre-training.

B10 · Annotation disagreement examples

Sentence fragment	Annotator A	Annotator B	Resolution
"killing at least 12 civilians"	CASUALTIES = "12 civilians"	CASUALTIES = "at least 12 civilians"	Include qualifier → B
"in eastern DRC"	REGION = "eastern DRC"	COUNTRY (dropped)	Eastern descriptor → REGION
"the al-shabaab"	ACTOR	ACTOR (lowercase tagged)	Case-insensitive → A
"the bus driver's family"	VICTIM	not annotated	Surface-form rule → VICTIM

IAA (Cohen's κ on a 200-doc pilot): 0.78 — substantial agreement; below 0.85 perfect agreement floor.

B11 · Error analysis — false-positive examples

Predicted span	Type	Why wrong
"this morning"	DATE	Vague temporal phrase, no anchor — should be untagged
"the convoy"	VICTIM	Generic NP, no specific victim — should be untagged
"Goma" (in "Goma football team")	CITY	Correct token, wrong context (sports team metaphor)
"Kinshasa region"	REGION	Should be DISTRICT (Kinshasa is a city-province)

Raising the WHEN threshold to 0.85 eliminates most "this morning"-class errors at cost of 1.2 F1 on legitimate DATE recall.

B12 · Error analysis — false-negative examples

Missed span	Type	Why missed
"Christian worshippers"	VICTIM	Religious-group framing absent from training
"were ambushed"	ACTION	Passive-voice action verb under-represented
"internally displaced schoolgirls"	VICTIM	Compositional, specific, rare in ACLED notes
"the bus driver's family"	VICTIM	Possessive construction not in templates

Future-work target: span-level CRF + boundary refinement (§7.5 high priority item 4).

End of backup deck

Main deck: slide 1–26. Backup: B1–B12. Speaker notes: `speaker_notes.md`. Q&A kit: `qa_kit.md`.